INTRODUCING

Kansas State University
College of Engineering
Computing and Information Sciences

HPC AND MULTITHREADED COMPUTING

TO MIDDLE SCHOOL GIRLS USING SCRATCH

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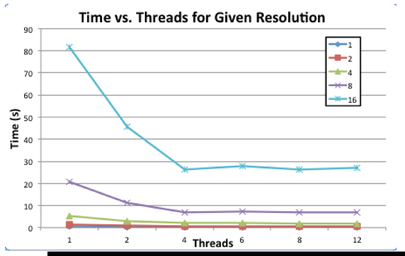


GOALS

-Introduce students to the concept of HPC in a brief (40 minute) presentation and hands on experiment

-Increase student awareness of the pervasiveness of computing

-Demonstrate the benefits of using computing (specifically HPC) When tackling large scale, complex problems

-Have students discover the trade-offs between model resolution and number of threads and the limitations of multithreading benefits due to system specifications

ACTIVITY

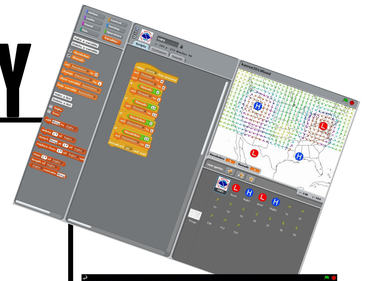
-Designed using the Scratch programming environment from MIT (scratch.mit.edu)

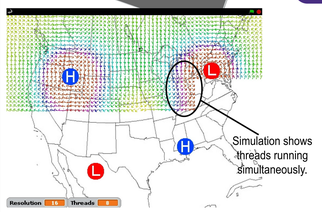
-Contains four different pressure centers

-Simulates the calculation of wind velocity based on pressure center locations

-Students can control the model resolution and number of threads used

-Simulation performance depends on hardware limitations





OUTCOMES

-After our presentation:
- -22 of 41 students showed interest in a job using HPC to solve problems
- -27 of 37 students felt they can learn how to write computer programs

-This activity will now be used in our “Introduction to Computing Science” class and other undergraduate classes to help introduce HPC

-We developed a framework in Scratch for teaching HPC concepts that can be easily adapted to simulate a multitude of real world problems

PROCEDURES

-Discuss parallelization are why it is needed for complex tasks such as weather modeling that are time sensitive

-Students worked in pairs to run the simulation activity, varying number of threads and model resolution and recorded the elapsed time

-Results were compared between groups to ensure data accuracy

-Data was summarized to show the benefits and limitations of performing multithreaded calculations on available hardware

Students performing activity

Recording results

Scott introducing activity

Tour of “Beocat” cluster

PRESENTATION

-Discussed everyday uses of HPC such as weather forecasting, and medical diagnosis

-Explained the basics of multithreaded computing and trade-offs that may be encountered

-Presented to 4 groups of around 20 middle school and high school girls

-Students toured an actual HPC cluster called “Beocat”

FUTURE WORK

-Improve the mathematical model used in the simulation to better simulate a real world computation

-Give the students more opportunity to customize the simulation in Scratch and learn programming fundamentals

-Improve our before and after survey questions to get better data on student understanding of and interest in HPC

-Schedule longer time periods with students to give us enough time to more fully explore HPC